

Rugved Zambare

Pune, India | +91 9373443632 | rugvedzambare05@gmail.com | github.com/rugvedzambare | linkedin.com/in/rugvedzambare

EDUCATION

Pimpri Chinchwad College of Engineering, Pune

2023 – 2027

B.Tech, Computer Engineering | Coursework: ML, Neural Networks, DSA, DBMS, Statistics

Vidya Valley North Point Junior College, Pune

2021 – 2023

Kamalnayan Bajaj School, Pune

2009 – 2021

TECHNICAL SKILLS

AI / ML: PyTorch, CNNs, Transfer Learning, scikit-learn, TrOCR, EasyOCR, Model Evaluation (F1, Precision/Recall, BLEU), AMP

LLMs: HuggingFace Transformers, Qwen3, Fine-Tuning, RLHF, Supervised Fine-Tuning, Streaming Inference, LLM Evaluation

MLOps & Cloud: Git, GitHub, GCP (Cloud Run), FastAPI, Flask, REST APIs, SQLite, Experiment Tracking, CI-style Pipelines

CV & Interfaces: OpenCV, MediaPipe, HSV Segmentation, NumPy, Pandas, Spotipy, OAuth 2.0 PKCE, Spotify Web API **CP:** LC 1650

Languages: Python, JavaScript, C++, C **Frontend:** HTML5, CSS3, JavaScript (ES6+)

PROJECTS

StackMind: Fine-Tuned LLM on Stack Overflow Dataset

Qwen3 · HuggingFace · FastAPI · GCP · Git

- **Reduced hallucination rate ~35%** vs Qwen3 baseline by fine-tuning on 500K+ Q&A pairs with HuggingFace Transformers, RLHF-aligned instruction formatting and automated BLEU/F1 evaluation in a Git-versioned MLOps pipeline.
- **Cut model iteration time ~40%** by replacing ad-hoc Colab notebooks with CI-style training loops, checkpoint versioning and Precision/Recall tracking across every training run.
- **Building a production streaming REST API** (FastAPI) on GCP Cloud Run; model card and weights to be published on HuggingFace Hub on release.

AI Assignment Grader: GPU OCR and Auto-Scoring Engine

PyTorch · TrOCR · EasyOCR · Flask · SQLite

- **7 ms per-assignment inference** on a 4 GB GTX 1650 using AMP mixed-precision and batched inference with a hybrid EasyOCR (line detection) and fine-tuned Microsoft TrOCR (handwriting) pipeline, tolerant of skewed phone photos via auto orientation detection.
- **Lifted OCR recall from ~33% to 100%** on noisy handwriting using a fragment-tolerant fuzzy matcher (difflib SequenceMatcher) that reconstructs OCR-shattered words by joining adjacent tokens.
- **Explainable grading engine** flags missing concepts, negation-detected wrong statements and OCR misreads without an LLM. ResNet18 score-bucket classifier at 90.8% accuracy, shipped end-to-end: OCR, scorer, SQLite, Flask, HTML/CSS/JS frontend.

NOVA: Gesture-Controlled Spotify Player

Python · OpenCV · MediaPipe · Spotipy · OAuth 2.0 PKCE

- **Touchless Spotify controller** using MediaPipe Hands (21 landmarks/hand) to classify real-time finger-state gestures via webcam for play, pause, skip and volume, with zero accidental triggers via a hold-frame debouncer and cooldown logic.
- **Full Spotify Connect integration** via Spotipy and the Spotify Web Playback SDK; registers as a named Connect device in the browser using OAuth 2.0 PKCE (client-side only), targeting it by name across search, playlists, library and queue.

EXPERIENCE

Google Developer Group on Campus (GDGOC) – PCCOE

Sep 2024 – Apr 2025

Design & AI/ML Team Member

- **Upskilled 100+ students** by co-facilitating PyTorch, Git and model evaluation workshops; 12+ student-led ML projects submitted at the semester showcase.
- **Delivered a production NFC/RFID Attendance System** with a REST API backend as a live club project, cutting manual attendance overhead ~80% across 3 departments.

Spectrum 2024 – PCCOE Annual Technical Fest

Feb 2024

Core Team Member

- **Executed a 500+ attendee event** with zero critical incidents, coordinated 8 volunteer teams and ran campaigns that exceeded registration targets by 20%.